Lecture 4

Crop science

Yield components

Yield

Crop yield is determined by plant growth and partitioning of biomass to marketable parts of plants

Growth and partitioning are affected by developmental stages of crops

Knowledge of physiological processes of growth, development, and partitioning into yield components is necessary for a basic understanding of maximizing crop yield

Yield parameters include leaf area, shoot, root, and total weights, or plant height

Yield components

Yield components can be divided into plant developmental stages of growth

Yield components are important in many crop research programs

Plant breeders improve yield by selecting components of yield such as seeds per plant or thousand grain weight

Components for grain yield of cereal crops, numbers of branches appear to be important in developing high-yielding cultivars

Grain yield in field crops is determined by components like ear per unit area, or ear weights

legumes are determined by numbers of pods per unit area, grains per pod, and dry weights of grain

Yield calculation

Yield = grain weight X number of grains per area

Eg yield of the maize plant with 100 grains per cob weighing 2 grams

100 grains X 2 grams = 200g X 11,111 = 2,222kg lima

200g X 44,444 plants = 88,888kg hectare

Dry matter production

Dry matter refers to the entire organic dry matter that is produced from essential activities of photosynthesis and protein metabolism

Organic dry matter includes glucose, starch, cellulose, amino acids, protein, and other organic components

Dry matter of plants refers to total biological yield which includes leaves, stalks, straw, roots, and other plant parts as well as economically useful parts such as grains, tubers, and fruit

For agricultural purposes, dry matter yields of many crops refer to aboveground (soil) grown plant parts but may contain belowground plant parts for some crops such as potatoes, cassava

Whole-plant aboveground biomass is the economically useful part of forage plants

Grain is the main useful part of cereals and legumes

Crop production can be measured as total biomass or economically useful plant parts

Total yield of plant material is known as biological yield

Harvest index is ratio of grain yield to biological yield

Eg 200g:1000g = 2:5 or 40%

Differences in translocation of photosynthate to grain may account for differences in harvest indexes

Harvest index

Efficiency of grain production in crop plants has also been frequently related to harvest index

Harvest index reflects partitioning of photosynthate between grain and vegetative plant parts

harvest index for a semidwarf wheat was 47 percent, compared to 40 percent for standard

Crop	Minimum	Maximum	Average
Millet	16	40	26
Sorghum	25	56	27
Maize	25	56	42
Rice	34	55	44
Wheat	35	49	41

Factors affecting productivity

Environment, management, and genotypes interact to determine overall yields of crops

Environmental factors, adequate moisture supply and optimum temperature and solar radiation are important in determining yield